Partial Fractions

Questions 1–10

Question 1. Express $\dfrac{2}{x(x^2+2)}$ in the form $\dfrac{A}{x}+\dfrac{Bx+C}{x^2+2}$.

Question 2. Express $\dfrac{3x^2-x+10}{(x+1)(x^2+1)}$ in the form $\dfrac{A}{x+1}+\dfrac{Bx+C}{x^2+1}$.

Question 3. A student is trying to express a rational function in partial fractions. Their working is shown below:

$$\frac{3x^2+5x+5}{(x+1)(x^2+2)} = \frac{A}{x+1} + \frac{B}{x^2+2}.$$

a) Explain the mistake made by the student.

b) Express $\dfrac{3x^2+5x+5}{(x+1)(x^2+2)}$ correctly as partial fractions.

Question 4. Write each of the following as partial fractions:

a) $\dfrac{x+5}{(x^2+1)(x+1)}$

b) $\dfrac{5x}{(2x+1)(x^2+1)}$

c) $\dfrac{x-2}{(x+3)(x^2+4)}$

d) $\dfrac{3x^2-6x+15}{(2x^2+1)(x-1)}$

e) $\dfrac{25x^2}{2x^3-3x^2+8x-12}$

f) $\dfrac{x^2}{x^3-3x^2+5x-15}$

g) $\dfrac{1}{(x^2+k)(x-1)}$, where k is a positive constant.

h) $\dfrac{k}{(x^2+k)(x-k)}$, where k is a positive constant.

Question 5. It is known that

$$\frac{6x^2+5x+6}{x^3+kx} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+k},$$

where k, A, B and C are positive integers. Determine the possible values of k. **The following questions require the use of Partial Fractions and A-Level Maths Integration Methods.**

Question 6. Find each of the following in simplified form:

a) $\displaystyle\int \frac{18}{(x-1)(x+2)(x-4)}\,dx$

b) $\displaystyle\int \frac{x-2}{(x-1)^2}\,dx$

c) $\displaystyle\int \frac{7x}{(2x-1)(x+3)^2}\,dx$

Question 7. Show that

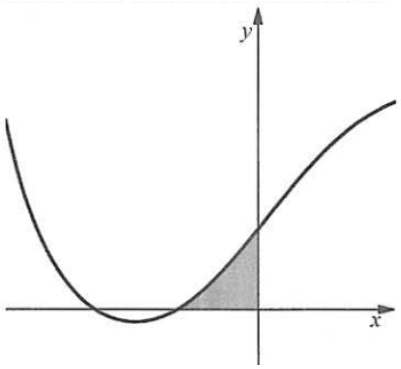
$$\int \frac{3x^2 + x + 4}{(x+1)(x^2+2)}\,dx = \ln\left((x+1)^2\sqrt{x^2+2}\right) + c.$$

Question 8. Show that

$$\int_1^4 \frac{x^2 - 4x + 15}{(x+1)(x^2+3)}\,dx = \ln\frac{3125}{722}.$$

Question 9. The diagram below shows part of the curve with equation

$$y = \frac{4x^2 + 6x + 2}{(x+2)(x^2+2)}, \qquad x \neq -2.$$



Find the exact value of the shaded area.

Question 10. Find the value of k such that

$$\int_4^6 \frac{(k-1)x^2 - 3kx - 4}{(x-3)(x^2+4)}\,dx = \ln\left(\frac{32}{3}\right).$$

Differentiation: Inverse Trigonometric Functions

Questions 1–26

Question 1. Show that

a) $\dfrac{d}{dx}(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}$

b) $\dfrac{d}{dx}(\arccos x) = -\dfrac{1}{\sqrt{1-x^2}}$

c) $\dfrac{d}{dx}(\arctan x) = \dfrac{1}{1+x^2}$

Question 2. Find the gradient of the tangent to the curve:

a) $y = \arcsin x$ at $x = \frac{1}{3}$

b) $y = \arccos x$ at $x = -\frac{1}{4}$

c) $y = \arctan x$ at $x = \frac{2}{5}$

The following questions use the derivatives of $\arcsin kx$, $\arccos kx$ and $\arctan kx$ and require no further Chain, Product or Quotient Rule.

Question 3. Differentiate each of the following:

a) $y = \arcsin 2x$

b) $y = \arccos 3x$

c) $y = \arctan 8x$

d) $y = \arcsin \dfrac{x}{3}$

e) $y = \arccos \dfrac{x}{5}$

f) $y = \arctan\left(-\dfrac{2}{3}x\right)$

Question 4. Show that

a) $\dfrac{d}{dx}\left(\arcsin \dfrac{x}{2}\right) = \dfrac{1}{\sqrt{4-x^2}}$

b) $\dfrac{d}{dx}\left(\arccos \dfrac{3x}{4}\right) = -\dfrac{3}{\sqrt{16-9x^2}}$

c) $\dfrac{d}{dx}\left(\arctan \dfrac{5x}{3}\right) = \dfrac{15}{9+25x^2}$

Question 5. Find the exact gradient of the curve $y = 8\arcsin\dfrac{x}{4}$ at the point with x-coordinate 2.

Question 6. Find the equation of the tangent to the curve $y = 3\arccos x$ when $x = \frac{1}{2}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

Question 7. Find the equation of the tangent to the curve $y = 2\arccos\dfrac{x}{3} - \arcsin\dfrac{x}{3}$ when $x = \dfrac{3}{\sqrt{2}}$, writing your answer in the form $y = ax + b$ where a and b are given in exact form.

Question 8. Find the equation of the tangent to the curve $y = 4\arctan\left(\dfrac{2x}{5}\right)$ when $x = \dfrac{5}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Find the equation of the normal to the curve $y = 4 - 3\arctan\dfrac{x}{4} - 2\arctan\dfrac{x}{3}$ when $x = 0$, writing your answer in the form $px + qy = r$ where p, q and r are integers.

Question 10. Given $f(x) = \arctan x$ and $g(x) = \arctan 3x$, solve the equation $f'(x) = g'(x)$.

Question 11. A curve has the equation $y = \arcsin\left(\frac{2}{3}x\right)$, where $-\frac{3}{2} \leq x \leq \frac{3}{2}$.

a) Show that the gradient of the curve is never equal to $\frac{1}{2}$.

The point P lies on the curve with x-coordinate $-\frac{3\sqrt{2}}{4}$. The normal to the curve at P passes through the y-axis at the point Q.

b) Show that the exact area of the triangle OPQ, where O is the origin, can be written as $\frac{3\sqrt{2}}{64}(9 + 2\pi)$ units^2.

Question 12. Show that the curve with equation $y = \arcsin x - \arccos x$ is increasing for $-1 < x < 1$.

Question 13. It is given that $y = 3\arctan 6x$.

a) Show that there are no points on the curve where the gradient is equal to 24.

b) State the maximum gradient of the curve.

Question 14. Find the exact coordinates of the single stationary point of the curve $y = \arctan 2x - x^2$. **Prerequisite knowledge for Q15: Binomial Series**

Question 15. It is given that $f(x) = \arctan 3x$.

a) Find $f'(x)$.

b) Assuming that x is small, use your answer to part (a) to express $f'(x)$ in ascending powers of x as far as the term in x^4.

c) Hence express $\arctan 3x$ in ascending powers of x as far as the term in x^5.

Question 16. It is given that $f(x) = p\arcsin x - qx$, where p and q are positive constants. Show that the curve $y = f(x)$ has precisely two stationary points when $p < q$.

Question 17. Find the values of x for which the curve with equation $y = 5x + \sqrt{21}\arccos x$ is increasing. Write your answer in set notation.

Question 18. Show that the curve with equation $y = 3\arctan\dfrac{x}{6} - 2\arctan\dfrac{x}{4}$ has a stationary point of inflection at the origin. **The following questions use the derivatives of** $\arcsin kx$**,** $\arccos kx$ **and** $\arctan kx$ **and use any differentiation knowledge from A-Level Maths.**

Question 19. Differentiate each of the following:

a) $y = \arcsin x^2$

b) $y = \arccos \sqrt{x}$

c) $y = \arctan \dfrac{1}{x}$

d) $y = \cos(\arcsin 2x)$ for $-\dfrac{1}{2} < x < \dfrac{1}{2}$

e) $y = \sin\left(\arccos \dfrac{x}{2}\right)$ for $-2 < x < 2$

f) $y = \cos(\arctan 4x)$ for $x \in \mathbb{R}$

Question 20. Find the equation of the tangent to the curve $y = (\arcsin x)^2$ when $x = \dfrac{\sqrt{3}}{2}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 21. Find the equation of the tangent to the curve $y = \arctan(e^x)$ when $x = -\frac{1}{2}\ln 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 22. Differentiate each of the following:

a) $y = x \arcsin x$

b) $y = x^2 \arccos 2x$

c) $y = x^3 \arctan(x^2)$

Question 23. It is given that $y = (x-5)\sqrt{-x^2+10x-24} + \sin^{-1}(x-5)$, where $4 \leq x \leq 6$. Show that $\dfrac{dy}{dx} = k\sqrt{-x^2+10x-24}$, where k is an integer to be found.

Question 24. It is given that

$$y = 3x\sqrt{4-x^2} + 12\tan^{-1}\left(\frac{x}{\sqrt{4-x^2}}\right), \qquad -2 < x < 2.$$

Show that $\dfrac{dy}{dx} = k\sqrt{4-x^2}$, where k is an integer to be found.

Question 25. It is given that a curve has equation $\arccos y - \arccos 3x = \dfrac{\pi}{3}$, where $-\dfrac{1}{6} \leq x \leq \dfrac{1}{3}$. Find the value of $\dfrac{dy}{dx}$ when $x = \dfrac{1}{6}$.

Question 26. Show that

a) $\dfrac{d}{dx}(\text{cosec}^{-1} x) = -\dfrac{1}{\sqrt{x^4 - x^2}}$

b) $\dfrac{d}{dx}(\sec^{-1} x) = \dfrac{1}{\sqrt{x^4 - x^2}}$

c) $\dfrac{d}{dx}(\cot^{-1} x) = -\dfrac{1}{1 + x^2}$

Integration: Inverse Trigonometric Functions

Questions 1–51

Question 1. Use the substitution $x = \sin u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c.$$

Question 2. Use the substitution $x = \cos u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = -\arccos x + c.$$

Question 3. Find the exact value of

$$\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}}\, dx.$$

Question 4. Use the substitution $x = \tan u$ to show that

$$\int \frac{1}{1+x^2}\, dx = \arctan x + c.$$

Question 5. Find the exact value of

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx.$$

Question 6. Find:

a) $\displaystyle\int \frac{2}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{5}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-9x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+4x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{1-2x^2}}\,dx$

f) $\displaystyle\int \frac{1}{1+8x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{1-5x^2}}\,dx$

h) $\displaystyle\int \frac{200}{1+10x^2}\,dx$

i) $\displaystyle\int \frac{4}{\sqrt{1-\frac{1}{4}x^2}}\,dx$

j) $\displaystyle\int \frac{18}{1+\frac{2}{3}x^2}\,dx$

Question 7. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{2}/8} \frac{1}{\sqrt{1-16x^2}}\,dx$

b) $\displaystyle\int_{1/(2\sqrt{3})}^{1/2} \frac{30}{\sqrt{1-3x^2}}\,dx$

Question 8. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{3}/6} \frac{1}{1+36x^2}\,dx$

b) $\displaystyle\int_{1/\sqrt{3}}^{1} \frac{12}{1+3x^2}\,dx$

Question 9.

a) Show that

$$\frac{1}{2+x^2} \equiv \frac{1}{2(1+(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{2+x^2}\,dx$.

Question 10.

a) Show that

$$\frac{1}{\sqrt{2-x^2}} \equiv \frac{1}{\sqrt{2}\sqrt{1-(kx)^2}},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{\sqrt{2-x^2}}\,dx$.

Question 11. Find:

a) $\displaystyle\int \frac{1}{4+x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

c) $\displaystyle\int \frac{1}{5+x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{8-x^2}}\,dx$

e) $\displaystyle\int \frac{4}{12+x^2}\,dx$

f) $\displaystyle\int \frac{10}{\sqrt{5-x^2}}\,dx$

g) $\displaystyle\int \frac{1}{2+4x^2}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{16-9x^2}}\,dx$

i) $\displaystyle\int \frac{2}{4+25x^2}\,dx$

j) $\displaystyle\int \frac{3}{\sqrt{12-\frac{1}{3}x^2}}\,dx$

Question 12. Find the exact values of:

a) $\displaystyle\int_{-\sqrt{3}/3}^{\sqrt{3}/3} \frac{1}{\sqrt{4-9x^2}}\,dx$

b) $\displaystyle\int_{0}^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100-16x^2}}\,dx$

Question 13. Find the exact values of:

a) $\displaystyle\int_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \frac{1}{25+81x^2}\,dx$

b) $\displaystyle\int_{0}^{3(\sqrt{2}-\sqrt{3/2})} \frac{9}{9+2x^2}\,dx$

Question 14. Given that a and b are constants, find:

a) $\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{a^2+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx$

d) $\displaystyle\int \frac{1}{a^2+b^2x^2}\,dx$

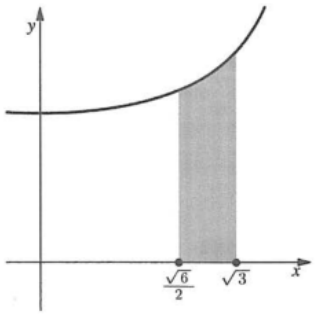
Question 15. Given that a and b are positive constants, find the exact value of

$$\int_0^{\sqrt{ab}} \frac{1}{\sqrt{b - \frac{1}{a}x^2}}\, dx.$$

Question 16. The curve shown in the diagram below has equation

$$y = \frac{6}{\sqrt{18 - 3x^2}}.$$

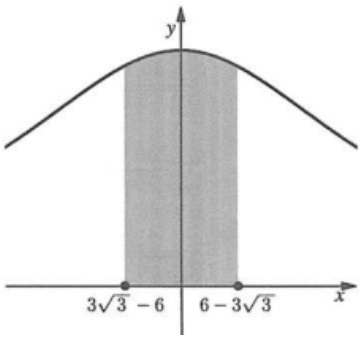
Find the exact area of the shaded region.



Question 17. The curve shown in the diagram below has equation

$$y = \frac{9}{36 + 4x^2}.$$

Find the exact area of the shaded region.



Question 18. It is given that

$$f(x) = \frac{19x^2 + 34}{(1 + x^2)(9 + 4x^2)}, \qquad x \in \mathbb{R}.$$

a) Write $f(x)$ in the form

$$\frac{Ax + B}{1 + x^2} + \frac{Cx + D}{9 + 4x^2},$$

where A, B, C and D are constants to be found.

b) Hence find $\int f(x)\,dx$.

Question 19. Find:

a) $\displaystyle\int \frac{107x^2 - 3}{(4 + 9x^2)(1 + 25x^2)}\, dx$

b) $\displaystyle\int \frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)}\, dx$

Question 20. Show that

$$\int_{1/2}^{\sqrt{3}/2} \frac{16x^2 + 5x + 4}{x(8x^2 + 2)}\, dx = a\pi + \ln b,$$

where a is a rational number and b is a positive integer to be found.

Question 21. Given that

$$\frac{1+x}{1+x^2} \equiv \frac{1}{1+x^2} + \frac{x}{1+x^2},$$

find $\displaystyle\int \frac{1+x}{1+x^2}\,dx$.

Question 22. Find:

a) $\displaystyle\int \frac{2x+1}{1+4x^2}\,dx$

b) $\displaystyle\int \frac{3+5x}{4+x^2}\,dx$

c) $\displaystyle\int \frac{2-3x}{9+4x^2}\,dx$

d) $\displaystyle\int \frac{5-4x}{25+16x^2}\,dx$

Question 23. Given

$$\frac{1+x}{\sqrt{1-x^2}} \equiv \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{1-x^2}}\,dx$.

Question 24. Find:

a) $\displaystyle\int \frac{3x-1}{\sqrt{1-9x^2}}\,dx$

b) $\displaystyle\int \frac{1+8x}{\sqrt{4-x^2}}\,dx$

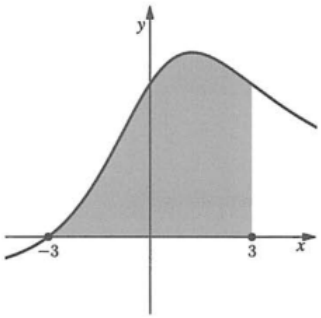
c) $\displaystyle\int \frac{6-x}{\sqrt{9-4x^2}}\,dx$

d) $\displaystyle\int \frac{10-9x}{\sqrt{4-25x^2}}\,dx$

Question 25. The curve shown in the diagram below has equation

$$y = \frac{3+x}{9+x^2}.$$

Find the exact area of the shaded region.



Question 26.

a) Find the general solution to the differential equation

$$\frac{dy}{dx} = \frac{x+1}{\sqrt{9-x^2}}.$$

b) Hence, find a particular solution such that $y = 3$ when $x = 0$.

Question 27.

a) Find $\displaystyle\int \frac{x}{\sqrt{25-x^2}}\,dx$.

b) Find $\displaystyle\int \frac{1}{\sqrt{25-x^2}}\,dx$.

c) It is given that

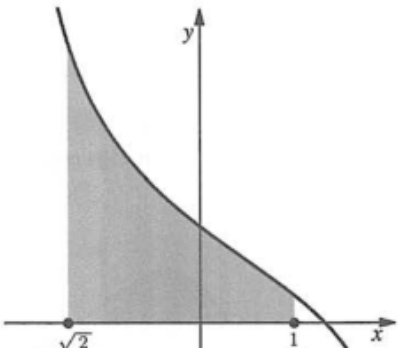
$$f(x) = \frac{5x+20}{\sqrt{25-x^2}}, \qquad |x| < 5.$$

Find the exact mean value of $f(x)$ over the interval $\left[-\frac{5}{2}, 0\right]$.

Question 28. The curve shown in the diagram below has equation

$$y = \frac{4-9x}{\sqrt{4-9x^2}}.$$

Find the exact area of the shaded region.



Question 29. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+9x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

f) $\displaystyle\int \frac{1}{4+x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{1-25x^2}}\,dx$

h) $\displaystyle\int \frac{1}{1+81x^2}\,dx$

i) $\displaystyle\int \frac{1}{\sqrt{4-9x^2}}\,dx$

j) $\displaystyle\int \frac{1}{9+4x^2}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{25-4x^2}}\,dx$

l) $\displaystyle\int \frac{1}{49+36x^2}\,dx$

m) $\displaystyle\int \frac{1}{\sqrt{16-3x^2}}\,dx$

n) $\displaystyle\int \frac{1}{4+5x^2}\,dx$

o) $\displaystyle\int \frac{1}{\sqrt{9-81x^2}}\,dx$

Question 30. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{1-x^2}\,dx$

b) $\displaystyle\int (1-x^2)^{3/2}\,dx$

Question 31. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1+x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(4+x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1+x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(9+4x^2)^3}\,dx$

Question 32. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{36-x^2}\,dx$

b) $\displaystyle\int (1-4x^2)^{3/2}\,dx$

Question 33. Complete the square for each of these:

a) $26 - 10x + x^2$

b) $101 + 20x + x^2$

c) $2 + 4x + 4x^2$

d) $5 - 12x + 9x^2$

e) $25 - 8x + x^2$

f) $29 + 8x + 4x^2$

Question 34.

a) Write $2 + 2x + x^2$ in the form $p + (x + q)^2$, where p and q are integers.

b) Hence, using the substitution $x + 1 = \tan u$, find $\displaystyle\int \frac{1}{2 + 2x + x^2}\, dx$.

Question 35. Find:

a) $\displaystyle\int \frac{1}{10-6x+x^2}\,dx$

b) $\displaystyle\int \frac{1}{65+16x+x^2}\,dx$

Question 36.

a) Write $10+18x+9x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $3x+3=\tan u$, find $\displaystyle\int \frac{1}{10+18x+9x^2}\,dx$.

Question 37. Find:

a) $\displaystyle\int \frac{1}{10+12x+4x^2}\,dx$

b) $\displaystyle\int \frac{1}{50-28x+4x^2}\,dx$

Question 38.

a) Write $13-6x+x^2$ in the form $p+(x+q)^2$ where p and q are integers.

b) Hence, using the substitution $\frac{x-3}{2}=\tan u$, find $\displaystyle\int \frac{1}{13-6x+x^2}\,dx$.

Question 39. Find:

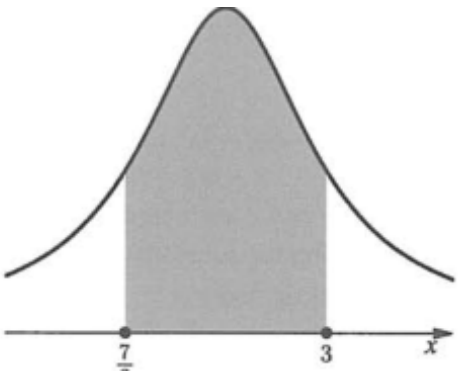
a) $\displaystyle\int \frac{1}{26+8x+2x^2}\,dx$

b) $\displaystyle\int \frac{6}{5+12x+8x^2}\,dx$

Question 40. The curve shown in the diagram below has equation

$$y = \frac{2}{65-48x+9x^2}.$$

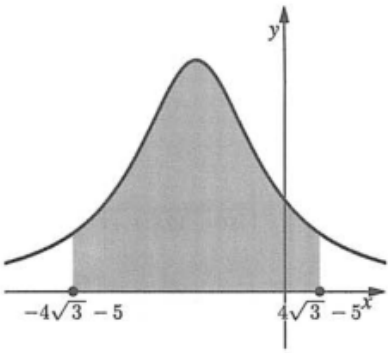
Find the exact area of the shaded region.



Question 41. The curve shown in the diagram below has equation

$$y = \frac{7}{41 + 10x + x^2}.$$

Find the exact area of the shaded region.



Question 42. Complete the square for each of these:

a) $-8 + 6x - x^2$
b) $-48 - 14x - x^2$
c) $-15 + 16x - 4x^2$
d) $-224 - 90x - 9x^2$
e) $-32 + 12x - x^2$
f) $-128 - 48x - 4x^2$

Question 43.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$ where p and q are integers.

b) Hence, using the substitution $x - 8 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-63 + 16x - x^2}}\, dx$.

Question 44. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-3 - 4x - x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-80 + 18x - x^2}}\, dx$

Question 45.

a) Write $-35 - 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $2x + 6 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-35 - 24x - 4x^2}}\, dx$.

Question 46. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-63 - 32x - 4x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-99 + 100x - 25x^2}}\, dx$

Question 47.

a) Write $-27 + 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}x - 2 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-27 + 24x - 4x^2}}\, dx$.

Question 48. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-11 - 36x - 9x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{3 + 10x - 25x^2}}\, dx$

Question 49. Show that

$$\int_2^{9/2} \frac{1}{\sqrt{21+4x-x^2}}\,dx = k\pi,$$

where k is a rational number to be found.

Question 50. A curve has equation

$$y = \frac{1}{\sqrt{400+x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, the y-axis and the line $x = 20$ is rotated through 2π radians about the x-axis.

Question 51. Show that

$$\int_{2/5}^{\infty} \frac{1}{4+25x^2}\,dx = k\pi,$$

where k is a rational number to be found.

Differentiation: Inverse Hyperbolic Functions

Questions 1–27

Question 1. Show that

$$\frac{d}{dx}(\operatorname{arsinh} x) = \frac{1}{\sqrt{x^2+1}}.$$

Question 2. Show that

$$\frac{d}{dx}(\operatorname{arcosh} x) = \frac{1}{\sqrt{x^2-1}}.$$

Question 3. Show that

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}.$$

Question 4. Differentiate each of the following:

a) $y = \operatorname{arsinh} 3x$
b) $y = \operatorname{arcosh} 2x$
c) $y = \operatorname{artanh} 5x$
d) $y = \operatorname{arsinh} \frac{1}{2}x$
e) $y = \operatorname{arcosh} \frac{x}{4}$
f) $y = \operatorname{artanh}\left(-\frac{x}{6}\right)$

Question 5. Differentiate each of the following:

a) $y = 2\,\text{arsinh}\,2x - 2\,\text{arcosh}\,4x$

b) $y = \frac{1}{3}\,\text{artanh}\,3x - \frac{2}{3}\,\text{artanh}\,6x$

Question 6. Find the equation of the tangent to the curve $y = 4\,\text{arsinh}\,x$ when $x = 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 7. Find the equation of the tangent to the curve $y = -6\operatorname{arcosh}\frac{x}{2}$ when $x = 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 8. Find the equation of the tangent to the curve $y = 10\operatorname{artanh}\frac{x}{5}$ when $x = 3$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Determine whether the curve with equation

$$y = \operatorname{arsinh} 4x - \operatorname{arcosh} 2x$$

is convex (concave upwards) or concave (concave downwards) when $x = 2$.

Question 10. It is given that the curve with equation

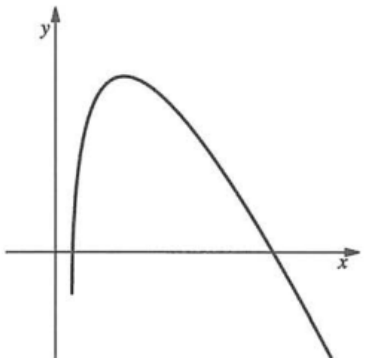
$$y = \sqrt{x^2 + 1} + \operatorname{arsinh} x$$

has a single stationary point. Find the exact coordinates of the stationary point.

Question 11. The graph of $y = f(x)$, where

$$f(x) = 12 \operatorname{arcosh} x - 3x, \qquad x \geq 1,$$

is shown below. Find the exact range of f, giving your answer in set notation.



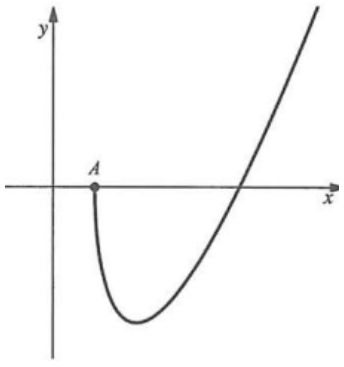
Question 12. The graph of $y = f(x)$ is shown below, where

$$f(x) = 2\sqrt{x^2 - 9} - 12 \operatorname{arcosh} \frac{x}{3}.$$

The point A lies on the curve and on the x-axis.

a) State the coordinates of the point A.

b) Find the range of f in set notation.



Question 13. It is given that the curve with equation

$$y = 4\,\text{artanh}\,x + \ln(x+1) - 6x$$

has two stationary points. It is also given that the curve has a single point of inflection.

a) Find the exact coordinates of both stationary points.

b) Find the exact x-coordinate of the point of inflection. Fully justify your answer.

Question 14. Differentiate each of the following:

a) $y = \text{arsinh}(\sqrt{x})$

b) $y = \text{arcosh}\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \text{artanh}(\sqrt[3]{x})$

Question 15. It is given that $x > 0$.

a) Show that $\dfrac{d}{dx}\left(\operatorname{arsinh}\dfrac{1}{x}\right) = -\dfrac{1}{x\sqrt{x^2+1}}$.

b) A curve has equation $y = \sqrt{x^2+1} - \operatorname{arsinh}\frac{1}{x}$. Show that the curve has no stationary points.

Question 16.

a) Show that

$$\frac{d}{dx}\left(\operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}\right) = \frac{2}{\sqrt{4x^2+1}}.$$

b) A curve has equation

$$y = \sqrt{4x^2+1} - \operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}.$$

Show that the curve has a single stationary point and find its exact coordinates.

Question 17. It is given that

$$f(x) = \operatorname{arcosh} 2x + \operatorname{arcosh} \frac{2}{x}.$$

For $x > 0$, find the exact coordinates of the stationary point of the curve $y = f(x)$.

Question 18. Differentiate each of the following:

a) $y = \operatorname{arsinh}(\cosh x)$
b) $y = \operatorname{arcosh}(\tanh x)$
c) $y = \operatorname{artanh}(\sinh x)$
d) $y = \operatorname{arsinh}(\tanh x)$
e) $y = \operatorname{arcosh}(\sinh x)$
f) $y = \operatorname{artanh}(\cosh x)$

Question 19. Differentiate each of the following:

a) $y = x^2 \operatorname{arsinh} x$

b) $y = \sinh x \operatorname{arsinh} x$

c) $y = \cosh x \operatorname{arcosh} x$

d) $y = \operatorname{arsinh} x \operatorname{arcosh} x$

e) $y = \operatorname{artanh} 2x \operatorname{artanh} 4x$

Question 20. Differentiate each of the following:

a) $y = \dfrac{x}{\operatorname{arsinh} x}$

b) $y = \dfrac{\operatorname{arsinh} x}{\operatorname{arcosh} x}$

c) $y = \dfrac{\operatorname{arcosh} 3x}{\operatorname{arsinh} 6x}$

d) $y = \dfrac{\operatorname{arsinh} 4x}{\operatorname{artanh} 8x}$

Question 21. It is given that

$$y = \sinh(2\operatorname{arcosh} x), \qquad x \geq 1.$$

Show that

$$(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 4y.$$

Question 22. It is given that

$$y = (\operatorname{arsinh} 4x)^2, \qquad x > 0.$$

Show that

$$(16x^2 + 1)\frac{d^2y}{dx^2} + 16x\frac{dy}{dx} = 32.$$

Question 23. Show that

$$\frac{d}{dx}(\operatorname{arcosech} x) = -\frac{1}{x\sqrt{x^2+1}}.$$

Question 24. Show that

$$\frac{d}{dx}(\operatorname{arcsech} x) = -\frac{1}{x\sqrt{1-x^2}}.$$

Question 25. Show that

$$\frac{d}{dx}(\operatorname{arcoth} x) = \frac{1}{1-x^2}.$$

Question 26. A curve has equation

$$y = x^2 \operatorname{arcosech} 2x, \qquad x > 0.$$

a) Differentiate y with respect to x.
b) Hence find the exact gradient of the curve when $x = \frac{1}{2}$.

Question 27. A curve has equation

$$y = \operatorname{arcsech} x^2 + \operatorname{arcoth} 3x,$$

where $-1 \leq x < -\frac{1}{3}$. Find the exact gradient of the curve when $x = -\frac{1}{2}$.

Integration: Inverse Hyperbolic Functions

Questions 1–57

Question 1. Use the substitution $x = \sinh u$ to show that

$$\int \frac{1}{\sqrt{x^2+1}}\,dx = \operatorname{arsinh} x + c.$$

Question 2. Use the substitution $x = \cosh u$ to show that

$$\int \frac{1}{\sqrt{x^2-1}}\,dx = \operatorname{arcosh} x + c.$$

Question 3. Use the substitution $x = \tanh u$ to show that

$$\int \frac{1}{1-x^2}\,dx = \operatorname{artanh} x + c.$$

Question 4. Find:

a) $\displaystyle\int \frac{4}{\sqrt{x^2-1}}\,dx$

b) $\displaystyle\int \frac{1}{2\sqrt{x^2+1}}\,dx$

c) $\displaystyle\int \frac{6}{1-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{4x^2-1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9x^2+1}}\,dx$

f) $\displaystyle\int \frac{1}{1-25x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{16x^2-1}}\,dx$

h) $\displaystyle\int \frac{\sqrt{8}}{\sqrt{2x^2+1}}\,dx$

i) $\displaystyle\int \frac{100}{1-\frac{1}{5}x^2}\,dx$

Question 5.

a) Show that

$$\frac{1}{15-x^2} \equiv \frac{1}{15(1-(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{15-x^2}\,dx$.

Question 6.

a) Show that

$$\frac{1}{\sqrt{x^2+3}} \equiv \frac{1}{\sqrt{3}\sqrt{(kx)^2+1}},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{\sqrt{x^2+3}}\,dx$.

Question 7. Find:

a) $\displaystyle\int \frac{1}{25-x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2+9}}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{x^2-4}}\,dx$

d) $\displaystyle\int \frac{1}{8-x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{x^2+12}}\,dx$

f) $\displaystyle\int \frac{1}{\sqrt{x^2-20}}\,dx$

g) $\displaystyle\int \frac{5}{10-x^2}\,dx$

h) $\displaystyle\int \frac{14}{\sqrt{x^2+7}}\,dx$

i) $\displaystyle\int \frac{18}{\sqrt{x^2-3}}\,dx$

j) $\displaystyle\int \frac{1}{9-4x^2}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{9x^2+4}}\,dx$

l) $\displaystyle\int \frac{1}{\sqrt{25x^2-16}}\,dx$

m) $\displaystyle\int \frac{15}{4-\frac{1}{4}x^2}\,dx$

n) $\displaystyle\int \frac{3}{\sqrt{16x^2+\frac{1}{9}}}\,dx$

o) $\displaystyle\int \frac{18}{\ldots}$

Question 8. Given that a and k are constants, find:

a) $\displaystyle\int \frac{1}{a^2-x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2+a^2}}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{k^2x^2-a^2}}\,dx$

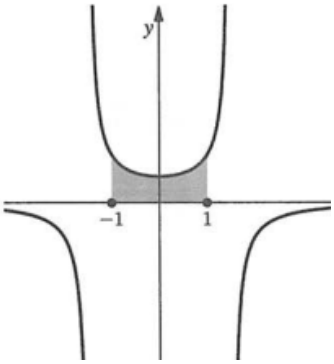
Question 9. Find the exact values of:

a) $\displaystyle\int_{-1/2}^{1/2} \frac{1}{1-x^2}\,dx$

b) $\displaystyle\int_{1}^{2} \frac{1}{\sqrt{4x^2-1}}\,dx$

c) $\displaystyle\int_{-5}^{5} \frac{30}{\sqrt{36x^2+25}}\,dx$

Question 10. The diagram below shows the curve $y = \frac{3}{9-4x^2}$. Show that the exact shaded area is $\frac{1}{2}\ln 5$.



Question 11. Show that

$$\int_{3/5}^{4/5} \left(\frac{5}{\sqrt{25x^2-4}} + \frac{31}{155x-92} \right) dx = \ln(6 - 2\sqrt{5} + 3\sqrt{3} - \sqrt{15}).$$

Question 12. Show that

$$\int_{1}^{3/2} \frac{4}{\sqrt{4x^2-1}}\, dx = \ln(a + b\sqrt{6} + c\sqrt{3} + d\sqrt{2}),$$

where a, b, c and d are integers to be found.

Question 13. It is given, for $x \in \mathbb{R}$, that

$$f(x) = \frac{19 - 39x^2}{(1 - x^2)(4 - 9x^2)}.$$

a) Write $f(x)$ in the form $\dfrac{Ax + B}{1 - x^2} + \dfrac{Cx + D}{4 - 9x^2}$, where A, B, C and D are constants to be found.

b) Show that $\displaystyle\int f(x)\,dx = p\,\text{artanh}(qx) + r\,\text{artanh}(sx) + c$, where p, q, r and s are constants to be found, and c is a constant of integration.

Question 14. Find:

a) $\displaystyle\int \frac{12x^2 + 1}{(1 - 9x^2)(1 - 16x^2)}\,dx$

b) $\displaystyle\int \frac{10x^2 + 47}{(1 - 4x^2)(25 - x^2)}\,dx$

c) $\displaystyle\int \frac{108x^4 + 12x^3 - 237x^2 - 27x + 25}{(1 - 9x^2)(9 - 4x^2)}\,dx$

Question 15. Given that

$$\frac{1-2x}{1-x^2} \equiv \frac{1}{1-x^2} - \frac{2x}{1-x^2},$$

find $\displaystyle\int \frac{1-2x}{1-x^2}\,dx$.

Question 16. Find:

a) $\displaystyle\int \frac{6x+1}{1-9x^2}\,dx$

b) $\displaystyle\int \frac{2-8x}{4-x^2}\,dx$

c) $\displaystyle\int \frac{8-3x}{16-9x^2}\,dx$

d) $\displaystyle\int \frac{10x-4}{25-100x^2}\,dx$

Question 17. Given

$$\frac{1+x}{\sqrt{x^2-1}} \equiv \frac{1}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{x^2-1}}\,dx$.

Question 18. Given

$$\frac{4x+1}{\sqrt{x^2+1}} \equiv \frac{4x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}},$$

find $\displaystyle\int \frac{4x+1}{\sqrt{x^2+1}}\,dx$.

Question 19. Find:

a) $\int \frac{2x-1}{\sqrt{4x^2-1}}\,dx$

b) $\int \frac{1-3x}{\sqrt{9x^2+1}}\,dx$

c) $\int \frac{4+6x}{\sqrt{9x^2-4}}\,dx$

d) $\int \frac{8x-3}{\sqrt{4x^2+36}}\,dx$

Question 20. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \frac{1}{\sqrt{x^2+1}}\,dx$

b) $\int \frac{1}{\sqrt{x^2-1}}\,dx$

c) $\int \frac{1}{1-x^2}\,dx$

d) $\int \frac{1}{\sqrt{4x^2+1}}\,dx$

e) $\int \frac{1}{\sqrt{x^2-9}}\,dx$

f) $\int \frac{1}{1-9x^2}\,dx$

g) $\int \frac{1}{\sqrt{x^2+25}}\,dx$

h) $\int \frac{1}{\sqrt{36x^2-1}}\,dx$

i) $\int \frac{1}{100-x^2}\,dx$

j) $\int \frac{1}{\sqrt{49x^2+1}}\,dx$

k) $\int \frac{1}{\sqrt{x^2-64}}\,dx$

l) $\int \frac{1}{1-144x^2}\,dx$

m) $\int \frac{1}{\sqrt{4x^2+9}}\,dx$

n) $\int \frac{1}{\sqrt{9x^2-16}}\,dx$

o) $\int \frac{1}{4-25x^2}\,dx$

p) $\int \frac{1}{\sqrt{36x^2+25}}\,dx$

Question 21. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 + 1)^{3/2}\, dx$

Question 22. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 - 1)^{3/2}\, dx$

Question 23. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1-x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(1-9x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1-x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(4-x^2)^3}\,dx$

Question 24. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \sqrt{25x^2+1}\,dx$

b) $\displaystyle\int (x^2+9)^{3/2}\,dx$

c) $\displaystyle\int \sqrt{x^2-36}\,dx$

d) $\displaystyle\int (81x^2-1)^{3/2}\,dx$

Question 25. Complete the square for each of these:

a) $x^2 + 4x + 3$
b) $x^2 - \frac{10}{3}x + \frac{34}{9}$
c) $4x^2 + 24x + 37$
d) $9x^2 - 18x + 8$
e) $x^2 + 6x + 18$
f) $9x^2 - 36x + 32$

Question 26.

a) Write $x^2 + 20x + 101$ in the form $(x + p)^2 + q$, where p and q are integers.
b) Hence, using the substitution $x + 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 + 20x + 101}}\, dx$.

Question 27.

a) Write $x^2 - 14x + 48$ in the form $(x + p)^2 + q$, where p and q are integers.

b) Hence, using the substitution $x - 7 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 - 14x + 48}}\, dx$.

Question 28. Find:

a) $\displaystyle\int \frac{1}{\sqrt{x^2 - 10x + 26}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2 + \frac{2}{3}x - \frac{8}{9}}}\, dx$

Question 29. Find the exact mean value of $f(x)$ over the interval $\left[-\frac{2}{5}, \frac{2}{5}\right]$, given

$$f(x) = \frac{1}{\sqrt{x^2 - \frac{4}{5}x + \frac{29}{25}}}, \qquad x \in \mathbb{R}.$$

Question 30.

a) Write $4x^2 - 40x + 101$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $2x - 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 40x + 101}}\, dx$.

Question 31.

a) Write $16x^2 + 96x + 143$ in the form $p(x + q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $4x + 12 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{16x^2 + 96x + 143}}\, dx$.

Question 32. Find:

a) $\displaystyle\int \frac{1}{\sqrt{9x^2 - 144x + 575}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{2x^2 + 12x + 19}}\, dx$

Question 33.

a) Write $4x^2 - 56x + 205$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}(x-7) = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 56x + 205}}\, dx$.

Question 34.

a) Write $3x^2 + 6x - 1$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{\sqrt{3}}{2}(x+1) = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{3x^2 + 6x - 1}}\, dx$.

Question 35. Find:

a) $\displaystyle\int \frac{1}{\sqrt{25x^2 - 100x + 91}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{5x^2 + 10\sqrt{5}x + 29}}\,dx$

Question 36. Show that

$$\int_{-1}^{1} \frac{1}{\sqrt{4x^2 + 8x + 13}}\,dx = \frac{1}{2}\ln k,$$

where k is a positive integer to be found.

Question 37. Complete the square for each of these:

a) $-3 - 4x - x^2$

b) $-24 + 10x - x^2$

c) $-15 - 16x - 4x^2$

d) $-143 + 72x - 9x^2$

e) $-45 - 14x - x^2$

f) $8 + 4x - 4x^2$

Question 38.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$, where p and q are integers.

b) Hence, using the substitution $x - 8 = \tanh u$, find $\displaystyle\int \frac{1}{(-63 + 16x - x^2)}\, dx$.

Question 39. Find:

a) $\displaystyle\int \frac{1}{(-99-20x-x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-168+26x-x^2)}\,dx$

Question 40.

a) Write $-35+24x-4x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $2x-6=\tanh u$, find $\displaystyle\int \frac{1}{(-35+24x-4x^2)}\,dx$.

Question 41. Find:

a) $\displaystyle\int \frac{1}{(-224-90x-9x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-24+50x-25x^2)}\,dx$

Question 42.

a) Write $11+10x-x^2$ in the form $p-(x+q)^2$, where p and q are integers.

b) Hence, using the substitution $\frac{x-5}{6}=\tanh u$, find $\displaystyle\int \frac{1}{(11+10x-x^2)}\,dx$.

Question 43. Find:

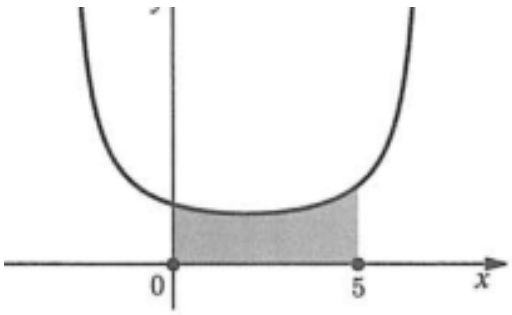
a) $\int \frac{1}{(91-18x-9x^2)}\,dx$

b) $\int \frac{1}{(-20-24x-4x^2)}\,dx$

Question 44. The curve shown in the diagram below has equation

$$y = \frac{5}{63+12x-3x^2}.$$

Find the exact area of the shaded region.



Question 45. A curve has equation

$$y = \frac{2}{\sqrt{-16 + 16x - 3x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, and the lines $x = 2$ and $x = 3$ is rotated through 2π radians about the x-axis.

Question 46. Use the substitution $x = \cosh u$ to find $\displaystyle\int \sqrt{x^2 - 1}\, dx$.

Question 47. Use the substitution $x = \sinh u$ to find $\int \sqrt{x^2 + 1}\, dx$.

Question 48. Use a suitable substitution to find $\int \sqrt{9x^2 - 1}\, dx$.

Question 49. Find the exact value of

$$\int_{12}^{13} \sqrt{x^2 - 144}\, dx.$$

Question 50. Show that

$$\int_0^2 \sqrt{25x^2 + 1}\, dx = \sqrt{101} + \frac{1}{10} \ln(10 + \sqrt{101}).$$

Question 51. Find:

a) $\displaystyle\int \sqrt{x^2 - 10x + 26}\, dx$

b) $\displaystyle\int \sqrt{x^2 + 6x + 8}\, dx$

Question 52. Show that

$$\int_3^5 \sqrt{x^2 - 8x + 17}\, dx = \sqrt{2} + \ln(1 + \sqrt{2}).$$

Question 53. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2-1}}\,dx$.

Question 54. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2+1}}\,dx$.

Question 55. Use a suitable substitution to find $\displaystyle\int \frac{x^3}{\sqrt{x^2+1}}\,dx$.

Question 56. Use the substitution $x = \sinh^2 u$ to find $\displaystyle\int \sqrt{\frac{x}{x+1}}\,dx$.

Question 57. Use the substitution $x = \cosh u$ to find $\int (x^2 - 1)^{3/2}\, dx$.

Integration: Substitutions Mix

Questions 1–8

Question 1. For each of the following, suggest a substitution for x that reduces the integral to $\int C\,d\theta$, where C is a constant.

a) $\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}\,dx$, for constant a

b) $\displaystyle\int \frac{1}{\sqrt{a^2+x^2}}\,dx$, for constant a

c) $\displaystyle\int \frac{1}{a^2-x^2}\,dx$, for constant a

d) $\displaystyle\int \frac{1}{\sqrt{x^2-a^2}}\,dx$, for constant a

e) $\displaystyle\int \frac{1}{a^2+x^2}\,dx$, for constant a

Question 2. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{\sqrt{9x^2+400}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-16x^2}}\,dx$

c) $\displaystyle\int \frac{1}{36-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{625x^2+1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{121x^2-16}}\,dx$

f) $\displaystyle\int \frac{1}{1+25x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{100x^2+9}}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{4-121x^2}}\,dx$

i) $\displaystyle\int \frac{16}{\sqrt{x^2-4}}\,dx$

j) $\displaystyle\int \frac{7}{144+49x^2}\,dx$

k) $\displaystyle\int \frac{24}{25-9x^2}\,dx$

l) $\displaystyle\int \frac{32}{\sqrt{81-2x^2}}\,dx$

m) $\displaystyle\int \frac{90}{\sqrt{9x^2-10}}\,dx$

n) $\displaystyle\int \frac{55}{50-121x^2}\,dx$

o) $\displaystyle\int \frac{36}{5+108x^2}\,dx$

Question 3. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\int \frac{3}{\sqrt{9x^2+4}}\,dx$

b) $\int \frac{14}{4+49x^2}\,dx$

c) $\int \frac{144}{9-16x^2}\,dx$

d) $\int \frac{8}{\sqrt{16x^2-9}}\,dx$

e) $\int \sqrt{25-4x^2}\,dx$

f) $\int \frac{24576}{(16-9x^2)^3}\,dx$

g) $\int (4-49x^2)^{3/2}\,dx$

h) $\int \sqrt{36x^2+25}\,dx$

i) $\int \frac{900000}{(100+9x^2)^3}\,dx$

j) $\int \frac{280}{(25-49x^2)^2}\,dx$

k) $\int \frac{250}{(25+2x^2)^2}\,dx$

l) $\int (3x^2-4)^{3/2}\,dx$

Question 4. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\int \frac{1}{\sqrt{1-(3x-1)^2}}\,dx$

b) $\int \frac{1}{\sqrt{(x-1)^2+9}}\,dx$

c) $\int \frac{1}{\sqrt{3-(2x+5)^2}}\,dx$

d) $\int \frac{1}{\sqrt{(3x-2)^2+8}}\,dx$

e) $\int \frac{1}{8+(4x-1)^2}\,dx$

f) $\int \frac{4}{11-(5x+3)^2}\,dx$

Question 5. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{4x^2+12x+10}\,dx$

b) $\displaystyle\int \frac{3}{\sqrt{3-4x-4x^2}}\,dx$

c) $\displaystyle\int \frac{2}{x^2-6x+14}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{7+4x-x^2}}\,dx$

e) $\displaystyle\int \frac{4}{3x^2+8x+6}\,dx$

f) $\displaystyle\int \frac{7}{\sqrt{5x^2+2x+12}}\,dx$

Question 6. By first suggesting a suitable substitution, show that:

a) $\displaystyle\int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\,dx = 2\arcsin\left(\frac{\sqrt{5}}{5}\right)$

b) $\displaystyle\int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\,dx = 8\ln\left(\frac{3+\sqrt{5}}{2}\right)$

c) $\displaystyle\int_{-3}^{-3/2} \frac{3}{45+24x+4x^2}\,dx = \frac{\pi}{8}$

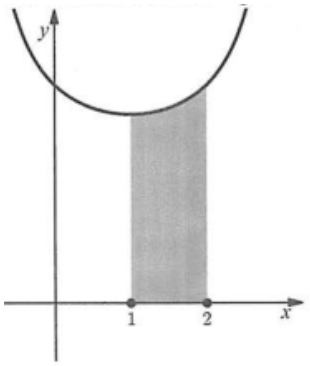
d) $\displaystyle\int_0^{3/2} \frac{2}{\sqrt{4x^2-12x+15}}\,dx = \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right)$

e) $\displaystyle\int_{-1}^{4} \sqrt{x^2+2x+10}\,dx = \frac{135}{32}+\frac{9}{2}\ln 2$

Question 7. The curve shown in the diagram below has equation

$$y = \frac{5}{\sqrt{3 + 2x - x^2}}.$$

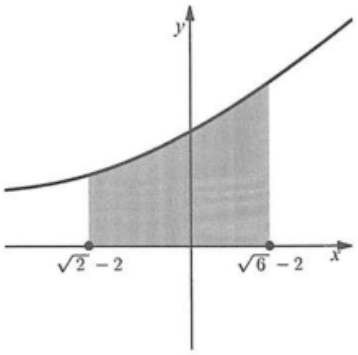
Find the exact area of the shaded region.



Question 8. The curve shown in the diagram below has equation

$$y = x + \frac{4}{6 + 4x + x^2}.$$

Find the exact area of the shaded region.



Integration: Advanced Methods (Trigonometric)

Questions 1–31

Question 1.

a) Use the substitution $x = \tan u$ to find $\displaystyle\int \frac{x^2}{1+x^2}\,dx$.

b) By writing $\dfrac{x^2}{1+x^2}$ in the form $A + \dfrac{B}{1+x^2}$, where A and B are integers, confirm your result from part a).

Question 2. Use the substitution $x = \tan u$, or otherwise, to find each of these.

a) $\displaystyle\int \frac{2x^2-1}{1+x^2}\,dx$

b) $\displaystyle\int \frac{5x^2+9}{1+x^2}\,dx$

Question 3. Find $\int \frac{1}{\sqrt{x(1+x)}}\,dx$ by:

a) using the substitution $\sqrt{x} = \tan u$

b) using the substitution $\sqrt{x} = u$

Question 4. Find $\int \frac{1}{(1+x^2)^{3/2}}\,dx$.

Question 5. Use the substitution $u = \sqrt{x^2 - 1}$ to find $\displaystyle\int \frac{1}{x\sqrt{x^2-1}}\,dx$.

Question 6.

a) Write $\sin^2\theta$ in terms of $\cos 2\theta$.

b) Use the substitution $x = \sin u$ to find $\displaystyle\int \frac{x^2}{\sqrt{1-x^2}}\,dx$.

Question 7. Use a suitable substitution to find each of these.

a) $\int \frac{x^2}{\sqrt{1-4x^2}}\,dx$

b) $\int \frac{x^2}{\sqrt{4-x^2}}\,dx$

c) $\int \frac{x^2}{\sqrt{4-2x^2}}\,dx$

Question 8. Use a suitable substitution to find each of these.

a) $\int \frac{2x^2+3}{\sqrt{1-x^2}}\,dx$

b) $\int \frac{3-x^2}{\sqrt{1-9x^2}}\,dx$

c) $\int \frac{3x^2+1}{\sqrt{1-3x^2}}\,dx$

Question 9. Find the exact value of $\displaystyle\int_{-1/4}^{1/4} \frac{8+8x^2}{\sqrt{1-4x^2}}\,dx$.

Question 10.

a) Show that $\dfrac{d}{dx}\left(3x\sqrt{1-9x^2} - \arccos(3x)\right) = k\sqrt{1-9x^2}$, where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_{1/6}^{\sqrt{3}/6} \sqrt{1-9x^2}\,dx$.

Question 11.

a) Show that

$$\frac{d}{dx}\left(2(x-5)\sqrt{11+10x-x^2}+72\arcsin\left(\frac{x-5}{6}\right)\right)=k\sqrt{11+10x-x^2},$$

where k is a positive integer to be found.

b) Hence find the exact value of $\displaystyle\int_5^8 \sqrt{11+10x-x^2}\,dx$.

Question 12.

a) Write $19+18x-x^2$ in the form $p-(x+q)^2$, where p and q are integers.

b) Using your answer to part a), use the substitution $\dfrac{x+q}{\sqrt{p}}=\sin u$ to show that

$$\int \sqrt{19+18x-x^2}\,dx=\frac{1}{2}(x-9)\sqrt{19+18x-x^2}+50\arcsin\left(\frac{x-9}{10}\right)+c.$$

Question 13. Use a suitable substitution to find $\int \sqrt{3 + 2x - x^2}\, dx$.

Question 14.

a) Show that $\sin 3A \equiv 3\sin A - 4\sin^3 A$.

b) Show that $\cos 3A \equiv 4\cos^3 A - 3\cos A$.

c) Use a suitable substitution to find $\int \frac{6x^3 - 4x + 1}{\sqrt{1 - x^2}}\, dx$.

Question 15.

a) Find $\dfrac{d}{dx}\left(\arctan\sqrt{\dfrac{x}{3}}\right)$.

b) Hence show that $\displaystyle\int_1^3 \frac{\sqrt{x}}{3x+x^2}\,dx = \frac{\pi}{6\sqrt{3}}$.

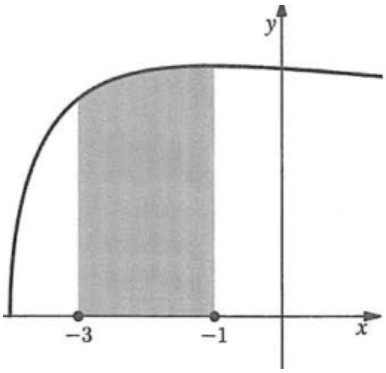
Question 16. Use the substitution $u^2 = x-3$, where $u > 0$, to find $\displaystyle\int \frac{\sqrt{x-3}}{x+6}\,dx$.

Question 17. Use the substitution $u^2 = x + 1$, where $u > 0$, to find $\displaystyle\int \frac{x}{\sqrt{x+1}(x+10)}\,dx$.

Question 18. The curve shown in the diagram below has equation

$$y = \frac{\sqrt{x+4}}{x+7}, \qquad x \geq -4.$$

The shaded region is the region bounded between the curve and the x-axis, and the lines $x = -3$ and $x = -1$. Show that the exact area of the shaded region is $\left(2\sqrt{3} - 2 - \frac{\pi}{2\sqrt{3}}\right)$ units^2.



Question 19. Use integration by parts to find:

a) $\int \arcsin x \, dx$

b) $\int \arccos x \, dx$

c) $\int \arctan x \, dx$

Question 20. Show that $\int_0^1 8 \arctan x \, dx = 2\pi - \ln p$, where p is a positive integer to be found.

Question 21. Find:

a) $\int \arcsin 2x \, dx$

b) $\int \arccos 3x \, dx$

c) $\int \arctan 4x \, dx$

Question 22. Find:

a) $\int 8x \arcsin(x^2) \, dx$

b) $\int e^{2x} \arccos(e^{2x}) \, dx$

c) $\int \sin x \arctan(\cos x) \, dx$

d) $\int \sec^2 x \arcsin(\tan x) \, dx$

e) $\int 3^x \arctan(3^x) \, dx$

Question 23. Find:

a) $\int x \arcsin x \, dx$

b) $\int x^2 \arccos x \, dx$

c) $\int x^3 \arctan x \, dx$

Question 24.

a) Write $\dfrac{10x^2}{1+x^2}$ in the form $A + \dfrac{B}{1+x^2}$.

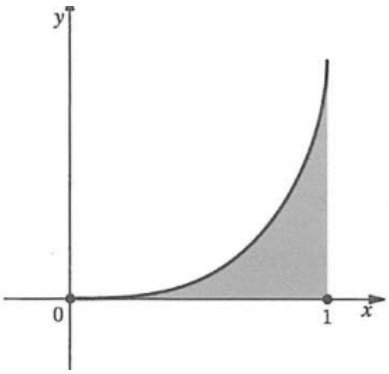
b) Find $\int 20x \arctan x \, dx$.

Question 25.

a) Write $\dfrac{15x^2}{4+x^2}$ in the form $A + \dfrac{B}{4+x^2}$.

b) Find $\displaystyle\int 30x \arctan\left(\frac{x}{2}\right)\, dx$.

Question 26. The curve shown in the diagram below has equation $y = 6x\arcsin(x^2)$, $0 \leq x \leq 1$. Find the exact area of the shaded region.



Question 27. Find:

a) $\displaystyle\int \frac{\arcsin x}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{\arccos x}{\sqrt{1-x^2}}\,dx$

c) $\displaystyle\int \frac{\arctan x}{1+x^2}\,dx$

Question 28. Show that $\displaystyle\int_{1/2}^{1/\sqrt{2}} \frac{(\arcsin x)^2}{\sqrt{1-x^2}}\,dx = k\pi^3$, where k is a rational number to be found.

Question 29. Find $\int \arctan x \ln(1+x^2)\,dx$.

Question 30. Find:

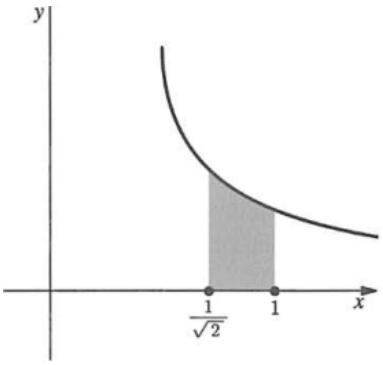
a) $\int \operatorname{arccosec} x\,dx, \quad x > 0$

b) $\int \operatorname{arcsec} x\,dx, \quad x > 0$

c) $\int \operatorname{arccot} x\,dx$

Question 31. The curve shown in the diagram below has equation $y = \operatorname{arccosec} 2x$ for $x > \frac{1}{2}$. The shaded region is the region bounded between the curve and the x-axis, and the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$. Show that the exact area of the shaded region is

$$\left(\frac{1}{2}\ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right) + \frac{\pi}{6} - \frac{\pi}{4\sqrt{2}}\right) \text{ units}^2.$$



Integration: Advanced Methods (Hyperbolic)

Questions 1–27

Question 1.

a) Show that $\operatorname{cosech} x = \dfrac{2e^x}{e^{2x} - 1}$.

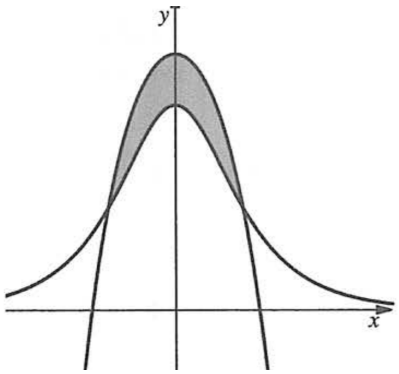
b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{cosech} x \, dx = -2 \operatorname{artanh}(e^x) + c$.

Question 2.

a) Show that $\operatorname{sech} x = \dfrac{2e^x}{e^{2x} + 1}$.

b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{sech} x \, dx = 2 \arctan(e^x) + c$.

Question 3. The shaded region shown in the diagram below is enclosed between the graphs of $y = 12\,\mathrm{sech}\,x$ and $y = 24 - 9\cosh x$. Find the exact shaded area in the form $p\ln(7+4\sqrt{3}) + q\sqrt{3} + r\pi$, where p, q and r are integers.



Question 4. Find:

a) $\displaystyle\int \tanh x\,dx$

b) $\displaystyle\int \coth x\,dx$

Question 5. Find:

a) $\int \operatorname{cosech} x \coth x \, dx$

b) $\int \operatorname{sech} x \tanh x \, dx$

Question 6.

a) By expressing $\sinh x$ and $\cosh x$ in terms of exponentials, prove that $\cosh^2 x - \sinh^2 x = 1$.

b) Find $\int \operatorname{cosech}^2 x \, dx$.

Question 7. Find the equation of the curve y given that its gradient function is:

a) $\dfrac{dy}{dx} = \operatorname{cosech} 3x \coth 3x$

b) $\dfrac{dy}{dx} = 3\operatorname{cosech}^2\left(\dfrac{x}{9}\right)$

c) $\dfrac{dy}{dx} = 6\operatorname{sech} 9x \tanh 9x$

d) $\dfrac{dy}{dx} = 10\coth 2x$

Question 8. Find $\displaystyle\int \frac{\cosh x}{\sinh^2 x + 1}\, dx$.

Question 9. Find $\displaystyle\int \frac{\cos x}{\sqrt{\sin^2 x + 1}}\, dx$.

Question 10.

a) Given that $u = \cosh^2 x$, show that $\dfrac{du}{dx} = \sinh 2x$.

b) Hence show that $\displaystyle\int \frac{\sinh 2x}{\cosh^4 x + 1}\, dx = \arctan(\cosh^2 x) + c$.

Question 11. Use integration by parts to find:

a) $\int \operatorname{arsinh} x \, dx$

b) $\int \operatorname{arcosh} x \, dx$

c) $\int \operatorname{artanh} x \, dx$

Question 12. Show that $\int_0^{1/3} 6 \operatorname{artanh} x \, dx = \ln\left(\frac{2^p}{3^q}\right)$, where p and q are positive integers.

Question 13. Show that $\displaystyle\int_0^{\frac{\tanh 4}{2}} 4\operatorname{artanh}(2x)\,dx = 8\tanh 4 + \ln|1-\tanh^2 4|$.

Question 14.

a) Find $\dfrac{d}{dx}(x\operatorname{arcosh} 3x)$.

b) Hence find $\displaystyle\int \operatorname{arcosh} 3x\,dx$.

Question 15. Find $\int x^2 \operatorname{arcosh} x \, dx$.

Question 16.

a) Find the values of A and B such that $\dfrac{2x^2}{1-x^2} = A + \dfrac{B}{1-x^2}$.

b) Hence find $\int 4x \operatorname{artanh} x \, dx$.

Question 17. Find:

a) $\displaystyle\int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$

b) $\displaystyle\int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,dx$

c) $\displaystyle\int \frac{\operatorname{artanh} x}{1-x^2}\,dx$

Question 18. Find the exact value of the following integral in terms of logarithms:

$$\int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,dx.$$

Question 19. Find the exact value of the following integral in terms of logarithms:

$$\int_{1/3}^{4} \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,dx.$$

Question 20.

a) Show that $\dfrac{10}{5\cosh x - 4\sinh x}$ can be written in the form $\dfrac{pe^x}{e^{2x}+q}$, where p and q are integers.

b) Hence show that $\displaystyle\int_0^{\ln 3} \frac{30}{5\cosh x - 4\sinh x}\,dx = 5\pi - 20\arctan\frac{1}{3}$.

Question 21. Use the substitution $u = \operatorname{artanh} x$ to find $\displaystyle\int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx$.

Question 22. A curve has equation $y = \operatorname{arcosh}\left(\dfrac{\sqrt{x}}{4}\right)$ for $x \geq \frac{1}{4}$.

a) Find $\dfrac{dy}{dx}$ and write it in a simplified form.

b) Hence show that $\displaystyle\int_{64}^{144} \frac{1}{\sqrt{x(x-16)}}\,dx = p\ln\left(\frac{q+p\sqrt{p}}{p+\sqrt{q}}\right)$, where p and q are positive integers.

Question 23. Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$.

Question 24.

a) Find $\dfrac{d}{dx}(\operatorname{arcosh} 2x)$.

b) Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}}\,dx$.

Question 25.

a) By using a double-angle identity, show that $\sinh(2\operatorname{arsinh} x) = 2x\sqrt{x^2+1}$.

b) Find $\displaystyle\int x \operatorname{arsinh} x \, dx$.

Question 26.

a) By writing in exponential form, show that $\displaystyle\int \operatorname{cosech} x \, dx = \ln\left|\tanh\frac{x}{2}\right| + c$.

b) i) By writing in exponential form, show that $\displaystyle\int \operatorname{sech} x \, dx = 2\arctan(e^x) + c$.

ii) Show that $e^x = \dfrac{1+\tanh(x/2)}{1-\tanh(x/2)}$.

iii) Hence show that $\displaystyle\int \operatorname{sech} x \, dx$ can also be written as $2\arctan\left(\tanh\frac{x}{2}\right) + c$.

Question 27.

a) By using exponential form, show that $\dfrac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$.

b) Hence find $\displaystyle\int x \operatorname{sech} x \tanh x \, dx$.

Improper Integrals

Questions 1–39

Question 1. Explain why each of the following are improper integrals.

a) $\int_1^\infty \sqrt{x}\,dx$

b) $\int_0^1 \frac{1}{3x^3}\,dx$

Question 2. Showing the limiting process used, find:

a) $\int_2^\infty \frac{1}{x^2}\,dx$

b) $\int_2^\infty \frac{1}{x^3}\,dx$

c) $\int_2^\infty \frac{1}{x^4}\,dx$

Question 3. Showing the limiting process used, find:

a) $\int_0^2 \frac{1}{x^2}\,dx$

b) $\int_0^2 \frac{1}{x^3}\,dx$

c) $\int_0^2 \frac{1}{x^4}\,dx$

Question 4. Showing the limiting process used, find:

a) $\int_0^9 \frac{1}{\sqrt{x}}\,dx$

b) $\int_4^\infty \frac{1}{\sqrt{x}}\,dx$

Question 5. Showing the limiting process used, find:

a) $\displaystyle\int_0^8 \frac{1}{x^{3/4}}\,dx$

b) $\displaystyle\int_8^\infty \frac{1}{x^{3/4}}\,dx$

Question 6. Showing the limiting process used, find:

a) $\displaystyle\int_4^\infty 192x^{-4}\,dx$

b) $\displaystyle\int_4^\infty (192x^{-4}+1)\,dx$

c) $\displaystyle\int_4^\infty 192x^{-4}(x+1)\,dx$

Question 7. Show clearly that $\displaystyle\int_0^4 \frac{1}{x^3\sqrt{x}}\,dx$ diverges.

Question 8. Show that $\displaystyle\int_0^{1/64} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt[3]{x}}\right) dx = \frac{5}{32}$.

Question 9. Amelia is attempting to evaluate the integral $\int_{-2}^{2} \frac{35}{x^6}\,dx$. Amelia's working is shown below:

$$\int_{-2}^{2} \frac{35}{x^6}\,dx = \int_{-2}^{2} 35x^{-6}\,dx = [-7x^{-5}]_{-2}^{2} = ((-7 \times 2^{-5}) - (-7 \times (-2)^{-5})) = -\frac{7}{32} - \frac{7}{32} = -\frac{7}{16}.$$

a) Explain the error in Amelia's method and what she should do instead.

b) Show clearly that $\int_{-2}^{2} \frac{35}{x^6}\,dx$ diverges.

Question 10. Find $\int_{-1}^{8} \frac{4}{x^{2/3}}\,dx$.

Question 11. Find $\int_{-32}^{1} \frac{100}{x^{4/5}}\,dx$. **The following questions require A-Level Maths Integration methods, including the Integrals of Standard Functions, Reversing the Chain Rule, Integration by Substitution, Integration by Parts, and Partial Fractions.**

Question 12. Explain why each of the following are improper integrals.

a) $\int_{4}^{6} \frac{1}{x-5}\,dx$

b) $\int_{0}^{1} \frac{e^x}{e^x-2}\,dx$

c) $\int_{0}^{2\pi/3} \frac{\sqrt{3}\sin x+\cos x}{\sin x-\sqrt{3}\cos x}\,dx$

d) $\int_{0}^{1} x^4 \ln x\,dx$

Question 13. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{x}\,dx$

b) $\displaystyle\int_0^2 \frac{1}{x}\,dx$

Question 14. Showing the limiting process used, find:

a) $\displaystyle\int_0^\infty \frac{1}{e^x}\,dx$

b) $\displaystyle\int_0^\infty \frac{1}{e^{2x}}\,dx$

Question 15. Show that $\displaystyle\int_7^\infty (x-3)^{-5/2}\,dx = \frac{1}{12}$.

Question 16. Showing the limiting process used, find $\displaystyle\int_{-2}^{62.5} \frac{2}{(2x+3)^6}\,dx$.

Question 17. Showing the limiting process used, find:

a) $\displaystyle\int_3^4 \frac{x}{x^2-9}\,dx$

b) $\displaystyle\int_3^4 \frac{x}{\sqrt{x^2-9}}\,dx$

Question 18. Showing the limiting process used, find:

a) $\displaystyle\int_0^\infty 4xe^{-x^2}\,dx$

b) $\displaystyle\int_{-\infty}^\infty 4xe^{-x^2}\,dx$

Question 19. Showing the limiting process used, find $\displaystyle\int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx$.

Question 20. Showing the limiting process used, find $\displaystyle\int_{0}^{\pi/2} \tan x\,dx$.

Question 21. Showing the limiting process used, find $\displaystyle\int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx$.

Question 22.

a) Show that $\displaystyle\lim_{a\to\infty}\left(\frac{6a-5}{3a+7}\right) = 2$.

b) Showing the limiting process used, find $\displaystyle\int_{4}^{\infty}\left(\frac{6}{6x-5} - \frac{3}{3x+7}\right)\,dx$.

Question 23. Show that $\displaystyle\int_1^{\infty} \frac{2}{x(3x+1)}\,dx = \ln k$, where k is a rational number to be found.

Question 24. Showing the limiting process used, find $\displaystyle\int_{-1}^{2} \frac{2-x}{\sqrt{x+1}}\,dx$.

Question 25. Showing the limiting process used, find $\displaystyle\int_{1/\pi}^{\infty} \frac{\sin(1/x)}{x^2}\,dx$.

Question 26. Using the substitution $u = 2 - x^3e^{-2x}$ and showing the limiting process used, show that

$$\int_{3/2}^{\infty} \frac{(3-2x)x^2}{x^3 - 2e^{2x}}\,dx = \ln\left(\frac{16e^3}{16e^3 - 27}\right).$$

The following questions require A-Level Further Maths Integration methods, including Inverse Trigonometric Functions, Hyperbolic Functions and Partial Fractions.

Question 27. By showing the limiting process used, show that $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \pi$.

Question 28. Showing the limiting process used, find $\int_{0}^{\infty} (\cosh x - \sinh x)\,dx$.

Question 29. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{4x^2-9}\,dx$

b) $\displaystyle\int_0^\infty \frac{1}{4x^2+9}\,dx$

Question 30. By showing the limiting process used, show that

$$\int_0^\infty \frac{12-6x}{(2x+3)(x^2+3)}\,dx = \ln k,$$

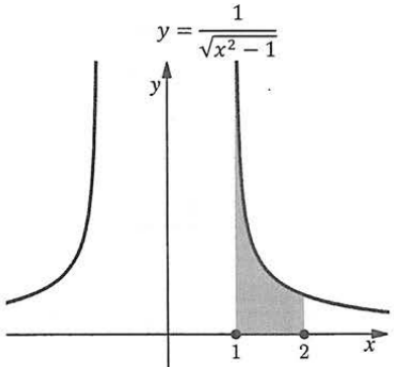
where k is a rational number to be found.

Question 31. By showing the limiting process used, show that

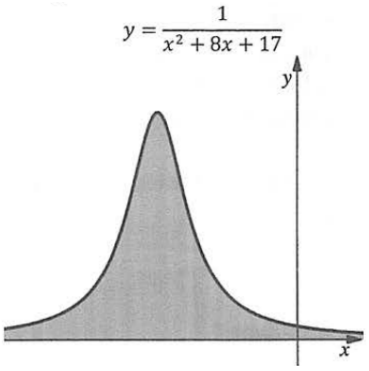
$$\int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,dx = 2 - k\pi,$$

where k is a rational number to be found.

Question 32. The diagram shown below shows the curve $y = \dfrac{1}{\sqrt{x^2-1}}$. Find the exact shaded area between the curve, the x-axis, the line $x = 1$ and the line $x = 2$.



Question 33. The diagram shown below shows the curve $y = \dfrac{1}{x^2 + 8x + 17}$. Find the exact total shaded area between the curve and the x-axis.



Using L'Hôpital's Rule

Question 34. Show that $\displaystyle\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = -4$.

Question 35. Showing the limiting process used, find $\int_0^{12} \ln x \, dx$.

Question 36. Showing the limiting process used, find $\int_0^{\infty} x^2 e^{-3x} \, dx$.

Question 37.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{\ln x}{x}\right)$.

b) Showing the limiting process used, find $\int_1^\infty \frac{\ln x}{x^2}\,dx$.

Question 38.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{x}{e^x}\right)$.

b) Showing the limiting process used, find $\int_{-\infty}^0 xe^x\,dx$.

Question 39. Show clearly that $\displaystyle\int_3^\infty \frac{\ln x - 1}{(\ln x)^2}\,dx$ diverges.

Mixed Integration Practice

Questions 1–8

Question 1. Find:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{x^2-1}}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{1+x^2}}\,dx$

e) $\displaystyle\int \frac{1}{1-x^2}\,dx$

Question 2.

a) Show that $\displaystyle\int_{\pi/4}^{\pi/2} 6\cot x\,dx = \ln k$, where k is a positive integer.

b) Show that $\displaystyle\int_{0}^{\pi/2} 6\cot x\,dx$ is undefined.

Question 3.

a) Show that the substitution $u = \tan x$ transforms $\displaystyle\int \frac{1}{25\cos^2 x + \sin^2 x}\,dx$ into $\displaystyle\int \frac{1}{25 + u^2}\,du$.

b) Hence show that $\displaystyle\int_0^{\arctan 5} \frac{1}{25\cos^2 x + \sin^2 x}\,dx = k\pi$, where k is a rational number to be found.

Question 4. Show that $\displaystyle\int_{\sqrt{\ln 2}}^{\sqrt{\ln 5}} \frac{x}{\sinh(x^2)}\,dx = \frac{1}{2}\ln 2$.

Question 5.

a) Find $\displaystyle\int \frac{1}{x^2 - 12x + 61}\, dx$.

b) Hence find the exact value of $\displaystyle\int_5^7 \left(3 - \frac{78}{x^2 - 12x + 61}\right) dx$.

c) The curve $y = 3 - \dfrac{78}{x^2 - 12x + 61}$ intersects the x-axis at the points P and Q.

a) Find the coordinates of P and Q.

b) The curve has a stationary point at the point R. Find the coordinates of R.

c) Find the exact area of the region between the curve, the x-axis, the line $x = 5$ and the line $x = 6$.

Question 6.

a) Find $\displaystyle\int \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx$.

b) Hence show that $\displaystyle\int_0^1 \frac{2x^3 + x^2 + x}{(x+1)(x^2+1)}\, dx = 2 - \ln 2 - \frac{\pi}{4}$.

Question 7. By using a suitable substitution, show that:

a) $\displaystyle\int_{-5/2}^{-3/2} \frac{1}{\sqrt{4x^2+20x+34}}\,dx = \frac{1}{2}\ln\left(\frac{2+\sqrt{13}}{3}\right)$

b) $\displaystyle\int_{1/2}^{5/4} \frac{1}{\sqrt{-4x^2+4x+8}}\,dx = \frac{\pi}{12}$

c) $\displaystyle\int_{(\sqrt{3}-3)/2}^{0} \frac{1}{\sqrt{4x^2+12x+18}}\,dx = \frac{\pi}{72}$

d) $\displaystyle\int_{3}^{4} \frac{1}{\sqrt{-4x^2+28x-40}}\,dx = \frac{1}{6}\ln 2$

Question 8. Show that $\displaystyle\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{12x^2}{1-x^4}\,dx = 3\ln(p+\sqrt{q}) + r\pi$, where p, q and r are integers to be found.